

Dhruv Varshney | 23EX10013

Bachelor of Science in Exploration Geophysics (4-Year), Dept. of Geology and Geophysics
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Education

Year	Degree/Exam	Institution	CGPA/Marks
2027	4YRS B.S	IIT Kharagpur	8.34/10
2023	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	91.2%
2021	Central Board of Secondary Education (CBSE)	Kendriya Vidyalaya IIT Kharagpur	97.6%

Internships

Analyst Programmer - Xceedance Consulting Private Limited

May '25 - Present

Extraction of Information from Insurance data using **LLM** and **Normalization** of insurance specific data.

- Generated Data using **genAI** and covering various types of data classes that can occur in insurance.
- Applied **Exploratory data analysis** on the insurance data and cleaned the data, enhancing data quality for downstream tasks.
- Normalized data** by standardizing and parsing names to ensure consistency in data format throughout the dataset.
- Built and deployed a **Few-Shot learning model** with the **Azure AI API** to accurately classify key entities in the data.

Vedvani: Transcribing Spoken Sanskrit Shlokas

May '24 - Nov '24

Under Prof. Pawan Goyal, IIT Kharagpur

Engineered a **speech-to-text pipeline** by scraping Sanskrit web text with **svaras** and training a **Wav2Vec2** model on the audio-text dataset

- Scraped a website using **BeautifulSoup** to gather a dataset of transcripts and corresponding audio files.
- Cleaned and preprocessed the dataset to ensure consistency and accurate alignment between audio and text.
- Trained a **Wav2Vec2 model** on the prepared dataset to perform speech-to-text transcription for Sanskrit.
- Evaluated model performance using **Word Error Rate (WER)** and **Character Error Rate (CER)** metrics, achieving improvements.

Projects

Home-coming Cloud Kitchen

May '25 - Present

- Built a fully functional and responsive static website for a local cloud kitchen using **HTML**, **CSS**, and **vanilla JavaScript**.
- Implemented a **dynamic menu system** that renders meal options in real-time based on user-selected dates and time slots.
- Designed an interactive order summary that recalculates subtotals, delivery fees, and totals as users add or adjust items.
- Integrated client-side form validation for accurate input and structured order details into a **JSON payload** for API simulation.
- Managed the project with **Git** and deployed the live site to **GitHub Pages**, showing end-to-end development proficiency.

Image-Filtering Web App

May '25 - Present

- Developed a fully responsive web app to apply various real-time image filters using **OpenCV** and **Flask**.
- Built interactive UI with **HTML**, **CSS**, **JS**, and **Fetch API** for dynamic previews and uploads, ensuring smooth interaction.
- Designed **RESTful Flask routes** for upload and processing; returned images via `send_file()` or `base64`.
- Utilized **OpenCV** for fast image transformations and optimized static delivery for repeated access, improving performance.

Sudoku Solver

Sept '24 - Nov '24

- Implemented a Sudoku puzzle solver using a **recursive backtracking algorithm** in **C++**.
- Integrated a **graphical user interface** using **SFML** to visualize the step-by-step solving process in real time.
- Enabled user input for custom puzzles and displayed solution path dynamically, demonstrating grasp of recursion and UI logic.
- Added functionality for highlighting conflicts and providing hints, further improving user experience and interactivity.

Awards and Achievements

- Ranked among the **top 6.4% in JEE Advanced 2023** and **top 1.9% in JEE Mains 2023**.
- Ranked among the **top 1% in WBJEE** (West Bengal Joint Entrance Exam) among more than 1 Lakh students.
- Achieved a **peak rating of 1200+ on Codeforces** demonstrating my Competitive programming skills.
- Solved 150+ Leetcode Problems** consistently improving my coding abilities and logical reasoning.
- Secured **4th place overall** in the **Open IIT Data Analytics Competition 2024**.

Skills and Expertise

- Languages:** C, C++, Python, HTML, CSS, Javascript, Bash, MySQL
- Libraries & Frameworks:** Pandas, NumPy, Matplotlib, scikit-learn, TensorFlow, BeautifulSoup4, spacy
- Tools & Platforms:** VS Code, Sublime Text, Git, Github, Jupyter Notebook, MS Office, Canva
- Concepts:** Large Language Models

Coursework Information

- Computer Science and Mathematics:** Programming and Data Structures (Lab and Theory), Linear Algebra, Advanced Calculus, Partial Differential Equation, Probability and Statistics, Transform Calculus
- Core and Electives:** Economics, Geophysical Signal Processing, Geophysical Field Theory, Introduction to Innovation and Entrepreneurship, Structural Geology, Petrology, Paleontology and Stratigraphy
- MOOCS:** **Machine Learning Specialization** by Andrew Ng (Coursera), Audio Course by Hugging Face, Matlab Onramp by Mathworks, Strivers A2Z Course

Positions of Responsibility

- Head of Design and Content Writing | Click Kgp:** Proficient in multiple photography styles including event, lifestyle, and wildlife, with advanced post-production skills.